

PVTRACE · DESIGN DECISION RECORD

PVTrace recommendations accepted

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DOCUMENT CONTROL

Document ID	PVT-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/pvtrace		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Items 10 and 11 decided by Amish on 2026-10-02 as recommended in PVT-DEC-001
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all recommendations and the changes made in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item with a recommendation is decided by Amish, 2026-09-25: go with recommendation. Items 10 and 11, which had no recommendation, were given recommendations in the design decisions register (PVT-DEC-001, items 5 and 4) and decided by Amish on 2026-10-02: "i approve your recommendations for all 555 open decisions."

■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." This record lists every PVTrace item that the decision closes, what changed in the repo because of it, and what stays open. The items come from the TRL 2 review note (docs/REVIEW.md , /populate session) and PVT-DDR-001. Where a recommendation offered several options, the recommended option is the decision. TRL 4 remains on hold by Amish's instruction, so nothing here is built, bought or tested.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation

#	Item	Decision	What changed in the repo
1	Load type	Capacitive load	Status wording only; already in PVT-PRC-001
2	Rating	100 V, 20 A, below the 120 V DC extra-low-voltage limit	Status wording only; R1, R2 unchanged
3	Load capacitance	6.6 mF, three 2200 µF 160 V capacitors	Status wording only; already in <code>bom/bom.csv</code> line 5 and the model
4	Isolation	Option B: digital isolator and isolated DC-DC converter	Status wording only; already in <code>bom/bom.csv</code> line 14 and the model
5	Display	2.8 in TFT plus phone export	Status wording only
6	Irradiance reference	Reference cell calibrated once against a pyranometer	Status wording only; the calibration itself is TRL 4 work, on hold
7	Controller	ESP32 display board	Status wording only
8	Grade thresholds	A 90 % or more of nameplate, B 80 to 90 %, C 70 to 80 %, reject below 70 % or with a visible safety defect, in a partner-editable table	R10 wording in PVT-REQ-001 v0.4 and PVT-CAL-001 v0.2
9	Budget	Raise <code>budget_usd</code> from \$150 to \$165	<code>project.yaml</code> <code>budget_usd</code> 150 to 165; README budget line; R17 in PVT-REQ-001 v0.4 changes from not met (\$163 against \$150) to met (\$164 against \$165, \$1 margin); PVT-PRB-001 v0.4 constraint
TRL 2 note item 11	<code>pitch</code> and <code>problem</code>	No change, as recommended	None
12	Heat and sun (R13)	Light grey case, charger with temperature cut-off and a display sun hood	New BOM line 15, printed PETG sun hood, 86 x 64 x 18 mm, \$1.00; hood added to <code>cad/src/model.py</code> (no clashes; top at Z 98, level with the isolator knob, so the R12 envelope is unchanged); STEP and STL re-exported; PVT-DWG-001 Rev P1 to P2; concept media refreshed with callout 15; PVT-CAL-001 v0.2: parts \$163 to \$164, mass 1.41 kg to 1.43 kg. R13 display readability stays not met on paper until a field check (TRL 4, on hold)
13	Case size	220 x 130 x 80 mm	Status wording only; already in the model and drawing

Items left open by this record

Table 2. Items left open by this record, both decided on 2026-10-02

#	Item	Decision
10	First co-design partner: a second-life panel refurbisher or recycler, or a TVET solar course	Decided by Amish, 2026-10-02: first candidate type to approach (not yet agreed), a second-life panel refurbisher or recycler that tests used modules in volume, with a TVET solar course as a good second partner for the teaching use (PVT-DEC-001, item 5)
11	R3 at worst-case capacitor tolerance (16.4 ms at -20 %): a fourth capacitor, selected capacitors of 6.4 mF or more, or relaxing the 20 ms bound	Decided by Amish, 2026-10-02: each capacitor is measured at build and a set is selected that gives at least 6.4 mF; the 20 ms bound is relaxed later only if TOPCon and HJT capacitance data justify it (PVT-DEC-001, item 4)

Cross-repo actions

None. CalRig's 10 to 40 °C range limit for the temperature probe (PVT-CAL-001 section 4) was a note, not a recommendation to change CalRig.

On hold (TRL 4)

Reference cell calibration, the display readability field check, any purchase, build or bench test. These follow from the decisions above but are TRL 4 work and stay on hold by Amish's instruction.

■ Consequences

- The budget is \$165 and the priced BOM is \$164, so R17 is met with a \$1 margin. Any later addition, such as a fourth capacitor for item 11 (about +\$6), would break it again.
- Requirements now stand at ten met, three not met (R3 at tolerance, R13 display readability, R16 out of scope), two at risk (R7, R8) and two not verifiable at TRL 3 (R6, R9).
- The sun hood adds about 15 g; mass is 1.43 kg against 1.5 kg (R12).
- `trl` and `trl_target` stay at 3.